MSG No. 51.304 P_Imma [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `.2/m.`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,3,5,7]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,4,6,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,11,13,15]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,12,14,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `.2/m.`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,3,5,7]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,4,6,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,11,13,15]`
4	$\left[0,\ \frac{1}{2},\ 0\right]$	`[10,12,14,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[1,6,10,13]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2,5,9,14]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[3,8,12,15]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[4,7,11,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,6,10,13]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,5,9,14]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,8,12,15]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[4,7,11,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4},\ 0,\ -z\right]$	`[2,3,5,8]`
3	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12,14,15]`
4	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,11,13,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `.2.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,3]`
2	$\left[\frac{1}{2},\ -y,\ 0\right]$	`[2,4]`
3	$[0,\ -y,\ 0]$	`[5,7]`
4	$\left[\frac{1}{2},\ y,\ 0\right]$	`[6,8]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,11]`
6	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[10,12]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[13,15]`
8	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[14,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[1,10]`
2	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2,9]`
3	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5,14]`
6	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[6,13]`
7	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[8,15]`

Table 8: Wyckoff site: `8h`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,7]`
2	$\left[x+\frac{1}{2},\ 0,\ -z\right]$	`[2,8]`
3	$[-x,\ 0,\ -z]$	`[3,5]`
4	$\left[\frac{1}{2}-x,\ 0,\ z\right]$	`[4,6]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,15]`
6	$\left[x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,16]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,13]`
8	$\left[-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,14]`

Table 9: Wyckoff site: `8i`, site symmetry: `m..`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,6]`
2	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[2,5]`
3	$\left[\frac{3}{4},\ y,\ -z\right]$	`[3,8]`
4	$\left[\frac{1}{4},\ -y,\ z\right]$	`[4,7]`
5	$\left[\frac{3}{4},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[10,13]`
7	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,16]`

continued ...

Table 9

No.	position	mapping
8	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12,15]`

Table 10: Wyckoff site: `16j`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$\left[\frac{1}{2}-x,\ y,\ z\right]$	`[6]`
7	$[x,\ -y,\ z]$	`[7]`
8	$\left[x+\frac{1}{2},\ y,\ -z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9]`
10	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[10]`
11	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[13]`
14	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[14]`
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[15]`
16	$\left[x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[16]`